

$$y = 2x + 3$$



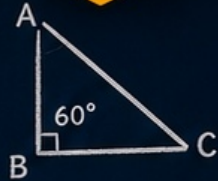
$$\frac{3}{4} + \frac{2}{4} = \frac{5}{4}$$

VOLUME
4

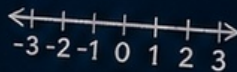
EMERGENCY SUB PLANS

for
MIDDLE SCHOOL MATH

KEEP LEARNING ON TRACK WHEN PLANS CHANGE



$$a^2 + b^2 = c^2$$



GRADES
6-8



NO-PREP
ACTIVITIES



READY-TO-USE
RESOURCES



PERFECT FOR
EMERGENCY
LESSONS



GREAT FOR
SUBSTITUTE
TEACHERS



PRINT OR
DIGITAL USE

RATIO & PROPORTIONS

Solve each problem.

- The ratio of boys to girls in a class is 3:5. If there are 24 girls, how many boys are in the class?



- Simplify the ratio 18:24.

SOLVE FOR x

Solve each equation.

- $2x + 5 = 17$

- $3(x - 4) = 21$

- $\frac{x}{4} + 3 = 8$

- $5x - 7 = 28$

PLOT & ANALYZE

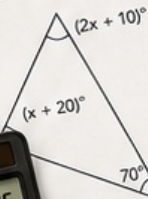
The graph shows the number of skateboards sold each week.



- How many skateboards were sold in Week 4?
- Between which weeks did sales decrease?

GEOMETRY

Find the value of x .



$x =$ _____

**MATH
EMERGENCY
KIT**

ORDER OF OPERATIONS

Evaluate each expression.

- $12 + 6 \times 2 - 4$
- $(18 - 6) \div 3 + 7$
- $2^4 + 3 \times (8 - 5)$
- $24 \div (2 + 2) + 5$

YOUR CLASSROOM LIFESAVER



MADE BY
TEACHERS
for
TEACHERS



ENGAGING
AND EFFECTIVE



SAVES YOU
TIME



STUDENTS STAY
ON TRACK



LOVED BY TEACHERS,
APPROVED BY PARENTS
AND STUDENTS!

Set this up while you are well

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a morning when you are not.

The one job you have today

Print pages 3–6 and leave them in a clearly labeled folder where a substitute will find them — top drawer, on the desk, or at the front office. Everything a sub needs is in those four pages.

You do not need to print the lessons ahead of time. The sub picks one on the day and copies it.

SETTING IT UP • FIFTEEN MINUTES, ONCE

- 1 **Print the whole pack once.** Put it in a folder or binder. Write your name and room number on the cover.
- 2 **Print pages 3–6 a second time** and clip them to the front. Those are the pages a substitute reads first.
- 3 **Tell your department chair where it lives.** On a real sick day you will not be answering the phone, so somebody else needs to know.
- 4 **Cross off lessons as you use them.** There is a tracker on page 4. Two years of absences will never repeat if you keep it current.
- 5 **Optional: leave a class roster** and a note about students who need a reader, extra time, or specific seating.

Why nothing here needs technology

On a sub day the Chromebook cart is often locked, the login server is down, or nobody knows the room code. Every lesson runs on paper and a pencil.

Calculators are optional throughout, and each lesson says where one actually helps.

Why it is not busy work

Every lesson targets a real Grade 6–8 standard and produces graded evidence you can put in a folder.

Away for a week? Running four of these in sequence is a defensible unit, not a holding pattern.

About the answer keys

Every key sits on the **last page of its lesson** and shows the **working**, not just the answer. That is what lets a sub who has not taught math in years respond when a student asks “*but why?*”

If you also own the other volumes

Volume 1, Volume 2, Volume 3 contain thirty further lessons and not one of them repeats a topic from this pack. Keep the folders side by side and a substitute can cover weeks without a student seeing the same sheet twice.

You do not need to be a math teacher

You have walked into a middle school math class. You may not have done algebra since high school. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to be able to do the math. Every lesson is a paper worksheet with a complete answer key that shows the working. Your job is to hand it out, keep the room settled, and check answers against the key. You are not expected to teach the content.

DO THESE FOUR THINGS

- 1 **Pick one lesson** from the table on the next page. Any of them works for Grades 6, 7 or 8. If you cannot decide, use Lesson 07.
- 2 **Copy the student pages only.** Each lesson tells you exactly which page numbers. One per student.
- 3 **Keep the teacher page and the answer key.** Do not copy those.
- 4 **Read the teacher page once** before the bell. Two minutes, and it includes the exact words you can use to start.

If a student says “we already did this”

Ask them to prove it by finishing the challenge section on the last page. If they genuinely have, switch them to a different lesson — there are ten, and they will not have done all of them.

If they finish early

Every lesson has a challenge task on its final student page. There is also a discussion question on the teacher page you can run with the whole class for the last five minutes.

What to leave behind

Collect the sheets. Fill in the handover note on the last page of this pack — ninety seconds, and it tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

Choose your lesson

PICK ANY ONE

10 OPTIONS

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	The Root of It	Square roots & perfect squares	★★	50	7–9
02	Never Ending	Rational & irrational numbers	★★	55	12–14
03	Money Over Time	Simple & compound interest	★★★	60	17–19
04	Same Shape, Different Size	Similar figures	★★	55	22–24
05	In and Out	Functions & function rules	★★	55	27–29
06	Two Lines, One Answer	Systems of equations	★★★	60	32–34
07	Many Sides	Angles in polygons	★	50	37–39
08	Scattered	Scatter plots & correlation	★★	55	42–44
09	Chance Twice	Compound events & tree diagrams	★★	55	47–49
10	Seeing in 3D	Cross-sections, plans & elevations	★★	55	52–54

Easiest to supervise

01, 07 and 10. Concrete and visual. A wrong answer usually looks wrong, so students catch themselves.

Best for a quiet class

02, 05 and 06. Steady practice with definite answers. Fast finishers will race — that is what the challenges are for.

Best for a restless class

03, 08 and 09. Bank accounts, scatter plots and dice. Noisier, but productively so.

Used-lesson tracker • for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Square roots			
02 — Rational & irrational			
03 — Interest			
04 — Similar figures			
05 — Functions			
06 — Systems			
07 — Polygon angles			
08 — Scatter plots			
09 — Compound events			
10 — 3D views			

The first ten minutes

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

"Come in and take a seat. Bags off the desks. You will not need a Chromebook today — everything is on paper."

THEN, ONCE THEY ARE SEATED

"Your teacher is out today. My name is _____. This is a real math lesson, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their own teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Settle, hand out sheets	Use the script above
5–10	Read the first panel together	Have a student read the worked example aloud
10–40	Students work through the tasks	Circulate. Check against the answer key
40–50	Challenge task for early finishers	Point them to the last student page
50–55	Collect sheets, pack up	Fill in the handover note

"I don't get it"

Almost always means *I don't know how to start*. Point at the worked example on the first page and ask them to copy its first line with their own numbers.

That single move unsticks most students without you needing to explain anything.

"Can I use a calculator?"

Unless the teacher page says otherwise, yes for the arithmetic and no for the reasoning tasks. Each teacher page states which.

If in doubt, allow it. A student doing the thinking with a calculator is better than one doing nothing.

Copying reminder

Student pages only. Each teacher page gives the exact range. The answer key is always the last page of the lesson — keep it.

The Root of It

2 MINUTES

Students learn the perfect squares up to 15^2 , take square roots of them, estimate the roots of numbers in between, and apply the idea to the side of a square garden and the size of a chessboard.

You do not need to know this subject

A square root undoes a squaring. $\sqrt{49}$ asks what number times itself gives 49. The perfect squares are printed on the student page and every answer is in the key.

Photocopy these pages only

Pages 7–9

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A calculator with a square-root key is useful for the estimating task.

WHAT TO SAY TO START

“Twelve times twelve is a hundred and forty-four. Everybody knows that one. Today we run it backwards — given the hundred and forty-four, find the twelve — and that reverse move turns out to be needed everywhere from garden fencing to television screens.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the perfect squares panel	Read it forwards and backwards
12–24	Task 1.1 — square and unsquare	Fast, confidence-building
24–34	Task 1.2 — trapping a root	Between which two whole numbers?
34–44	Task 1.3 — real squares	Gardens and screens
44–50	Challenge, collect sheets	Discussion question below

Questions students will ask

“What does the $\sqrt{}$ sign mean?”

It asks which number, multiplied by itself, gives what is underneath. $\sqrt{25}$ is 5 because $5 \times 5 = 25$.

“Is $\sqrt{2}$ a whole number?”

No, and it never becomes one however far you go. It is about 1.414. Lesson 02 in this volume is entirely about numbers like that.

“Why does $\sqrt{90}$ sit between 9 and 10?”

Because $9^2 = 81$ and $10^2 = 100$, and 90 sits between them. Trapping a root between two perfect squares is the whole method in Task 1.2.

If they finish early

Task 1.5 asks for the side of a square whose area is 200 m^2 .

There is no whole-number answer, and deciding what to round to is the interesting part.

Curriculum focus • for the regular teacher’s records

Grade 8: evaluating square roots of small perfect squares, and using rational approximations of irrational numbers to locate them on a number line.

Discussion question for the last five minutes: “Why does every positive number have two square roots, and why do we usually only write one?”

Forwards and backwards

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Squaring, and undoing it

THE PERFECT SQUARES

$1^2=1$	$6^2=36$	$11^2=121$
$2^2=4$	$7^2=49$	$12^2=144$
$3^2=9$	$8^2=64$	$13^2=169$
$4^2=16$	$9^2=81$	$14^2=196$
$5^2=25$	$10^2=100$	$15^2=225$

$$\sqrt{49} = 7$$

because $7 \times 7 = 49$

A square root undoes a squaring, the way dividing undoes multiplying.

$$7^2 = 49 \quad \text{and} \quad \sqrt{49} = 7$$

TRAPPING A ROOT

$$\sqrt{90} = ?$$

$$81 < 90 < 100$$

$$\sqrt{81} = 9, \quad \sqrt{100} = 10$$

so $\sqrt{90}$ is between **9 and 10**, nearer 10 (it is 9.49)

Task 1.1 • Square it, then unsquare it

DO

#	EXPRESSION	ANSWER	#	EXPRESSION	ANSWER
a	7^2		e	$\sqrt{1}$	
b	12^2		f	$\sqrt{0}$	
c	$\sqrt{49}$		g	$\sqrt{196}$	
d	$\sqrt{144}$		h	$\sqrt{25} + \sqrt{16}$	

In **h**, is $\sqrt{25} + \sqrt{16}$ the same as $\sqrt{41}$? Work both out. _____

Task 1.2 • Trap it between two whole numbers

ESTIMATE

#	ROOT	BETWEEN	AND	NEARER TO
a	$\sqrt{90}$			
b	$\sqrt{40}$			
c	$\sqrt{10}$			
d	$\sqrt{150}$			
e	$\sqrt{2}$			

Gardens, boards and claims

KEY SKILL

18 MIN

Task 1.3 • Real squares

APPLY

a. A square garden has an area of 169 m^2 . How long is one side, and how much fence goes all the way round?

b. A chessboard has 64 small squares in a square arrangement. How many along one edge?

c. A square photo has an area of 225 cm^2 . It is put in a frame 2 cm wide all round. What is the area of the framed picture?

d. A square car park holds 400 m^2 . A second one has sides twice as long. What is its area?

Task 1.4 • True or false?

JUDGE

Write true or false and give a reason or a counter-example.

#	STATEMENT	T / F	REASON
a	$\sqrt{36} = 6$		
b	Squaring a number always makes it bigger.		
c	$\sqrt{100} = 50$		
d	If a square has area 20 m^2 its side is a whole number.		
e	$\sqrt{9} \times \sqrt{4} = \sqrt{36}$		

Errors and edges

STRETCH

16 MIN

Task 1.5 • Find the error

DETECTIVE

Ada

$$\sqrt{64} = 32$$

What went wrong?

Correct answer: _____

Ben

$$\begin{aligned} & \sqrt{25} + \sqrt{16} \\ &= \sqrt{41} = 6.4 \end{aligned}$$

What went wrong?

Correct answer: _____

Cal

$$\begin{aligned} & \sqrt{2} = 1.41, \\ & \text{so } 1.41 \times 1.41 \\ & \text{is exactly } 2 \end{aligned}$$

What went wrong?

Correct answer: _____

Task 1.6 • Think it through

CHALLENGE

a. A square field has an area of 200 m^2 . How long is a side? Give your answer to one decimal place and say how you got there.

b. What is $5^2 \times 5^2$? Is it 5^4 or 25^2 , or both? _____

c. $(-6) \times (-6) = 36$. So does 36 have two square roots? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

The Root of It

ANSWERS

Task 1.1 • Square and unsquare

- | | |
|----------------------|----------------------|
| a. $7^2 = 49$ | e. $\sqrt{1} = 1$ |
| b. $12^2 = 144$ | f. $\sqrt{0} = 0$ |
| c. $\sqrt{49} = 7$ | g. $\sqrt{196} = 14$ |
| d. $\sqrt{144} = 12$ | h. $5 + 4 = 9$ |

h is the important one. $\sqrt{25} + \sqrt{16} = 9$, but $\sqrt{41} \approx 6.4$. They are nowhere near each other. You cannot add underneath a root sign, and this is the single most common misconception in the topic.

Task 1.2 • Trapping a root

- | | | |
|-----------------|-------------------|-------------------------------------|
| a. $\sqrt{90}$ | $81 < 90 < 100$ | 9 and 10 , nearer 10 |
| b. $\sqrt{40}$ | $36 < 40 < 49$ | 6 and 7 , nearer 6 |
| c. $\sqrt{10}$ | $9 < 10 < 16$ | 3 and 4 , nearer 3 |
| d. $\sqrt{150}$ | $144 < 150 < 169$ | 12 and 13 , nearer 12 |
| e. $\sqrt{2}$ | $1 < 2 < 4$ | 1 and 2 , nearer 1 |

Actual values: 9.49, 6.32, 3.16, 12.25, 1.41. Every estimate above is right, and none of them needed a calculator.

Task 1.3 • Real squares

- a. $\sqrt{169} = 13$ m per side, fence = $4 \times 13 = 52$ m
 b. $\sqrt{64} = 8$ squares along each edge
 c. $\sqrt{225} = 15$ cm; framed square is $15 + 2 + 2 = 19$ cm $\rightarrow 361$ cm²
 d. $\sqrt{400} = 20$ m; double is 40 m $\rightarrow 40^2 = 1,600$ m²

c catches people who add 2 to the side instead of 2 to each end. d is the one worth discussing: doubling the side did not double the area, it *quadrupled* it. Ask why before telling them.

Task 1.4 • True or false?

- a. **true** $6 \times 6 = 36$
 b. **false** $0.5^2 = 0.25$, smaller
 c. **false** $\sqrt{100} = 10$; 50 is half
 d. **false** $\sqrt{20} \approx 4.47$
 e. **true** $3 \times 2 = 6 = \sqrt{36}$

b and c are the two to dwell on. In c the student has halved rather than rooted, which works for 100 by coincidence of nothing at all. e is true and shows that roots *do* split over multiplication even though they never split over addition — compare it with Task 1.1h.

Task 1.5 • The three errors

Ada — 8. She halved 64. A root is not a half; $8 \times 8 = 64$.

Ben — 9. Roots cannot be added underneath the sign. Work each one out first: $5 + 4$.

Cal — not exactly. $1.41 \times 1.41 = 1.9881$, which is close but not 2. No decimal ever lands exactly on $\sqrt{2}$, which is what the next lesson is about.

Task 1.6 • Challenge

a. About 14.1 m. $14^2 = 196$ and $15^2 = 225$, so the side is just over 14. Trying 14.1 gives 198.81 and 14.2 gives 201.64, so 14.1 is the closer. Accept 14.1 or a well-argued 14.14.

b. Both. $5^2 \times 5^2 = 25 \times 25 = 625$, which is 5^4 and also 25^2 . Two different-looking expressions, one number.

c. Yes — 6 and -6. Every positive number has two square roots, one positive and one negative. The $\sqrt{}$ sign means specifically the positive one, which is a convention rather than a fact, and students who spot the missing half are right.

Never Ending

2 MINUTES

Students convert fractions to decimals and back, sort numbers into terminating, repeating and never-repeating, and find out why $\frac{1}{3}$ goes on forever while $\frac{1}{4}$ stops after two places. It ends with the fact that π has been calculated to trillions of digits and has never once repeated.

You do not need to know this subject

A number is rational if it can be written as one whole number over another. That single test settles every question on the sheet, and the key applies it to each one.

Photocopy these pages only

Pages 12–14

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A calculator is useful for the dividing.

WHAT TO SAY TO START

“A quarter is 0.25 and it stops. A third is 0.333 and it never stops. Both are perfectly ordinary fractions, so why does one of them behave like that? Today you find out, and then you meet numbers that are far stranger than either.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the rational panel	Rational means it can be a fraction
14–26	Task 2.1 — fractions to decimals	Just divide
26–36	Task 2.2 — sorting them	Three boxes
36–48	Task 2.3 and 2.4 — decimals back to fractions, and π	The interesting part
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“What does rational mean?”

That it can be written as one whole number divided by another. It has nothing to do with being sensible — the word comes from *ratio*.

“Is 0.333... rational?”

Yes. It is exactly $\frac{1}{3}$, which is a fraction, so it qualifies. Repeating forever does not stop a number being rational.

“So what is irrational?”

A decimal that never ends *and* never falls into a repeating pattern. $\sqrt{2}$ and π are the two students will meet. They cannot be written as any fraction at all.

If they finish early

Task 2.6 asks whether 0.999... equals 1. It does, exactly, and the argument for it is short enough to fit on the board and startling enough to be remembered.

Curriculum focus • for the regular teacher’s records

Grade 8: knowing that numbers that are not rational are called irrational, converting between fractions and decimals, and understanding that every rational number has a decimal expansion which terminates or repeats. Discussion question for the last five minutes: *“Can a number go on forever and still be exactly equal to a simple fraction?”*

Stops, repeats, or neither

UNDERSTAND

18 MIN

NAME _____

CLASS _____

DATE _____

Three kinds of decimal, two kinds of number

RATIONAL

can be written as
**one whole number over
another**

 $\frac{1}{2}$ 0.75 7 -5 0.333...

Decimal **stops** or **repeats**.

IRRATIONAL

cannot be written as
any fraction at all

 $\sqrt{2} = 1.41421356...$
 $\pi = 3.14159265...$

Never stops.
Never repeats.

WHY $\frac{1}{4}$ STOPS

$$\frac{1}{4} = 1 \div 4 = 0.25$$

4 divides into 100 exactly,
so the division finishes.

$$\frac{1}{3} = 1 \div 3 = 0.333...$$

3 never divides 10, 100 or
1000 exactly, so it never
finishes.

Task 2.1 • Fraction to decimal

DO

Divide the top by the bottom. If it repeats, write the first six digits and three dots.

#	FRACTION	DECIMAL	#	FRACTION	DECIMAL
a	1/4		e	5/8	
b	1/3		f	1/6	
c	3/8		g	7/20	
d	2/9		h	1/7	

Task 2.2 • Sort them

SORT

Put each number in exactly one column.

0.5 $\frac{1}{3}$ $\sqrt{9}$ $\sqrt{2}$ π 0.7 -4 $2\frac{1}{2}$ 0.161616... $\sqrt{16}$ 1.010010001... 22/7

DECIMAL STOPS

DECIMAL REPEATS FOREVER

NEVER STOPS, NEVER REPEATS

Which of your three columns holds the **irrational** numbers? _____

Back the other way

KEY SKILL

20 MIN

Task 2.3 • Decimal back to fraction

DO

Write each one as a fraction in its simplest form.

#	DECIMAL	FRACTION	#	DECIMAL	FRACTION
a	0.5		e	0.6	
b	0.25		f	0.125	
c	0.75		g	0.04	
d	0.2		h	1.5	

Task 2.4 • Where does it sit?

ORDER

Put these six numbers in order, smallest first. You will need a decimal value for each.

 $\sqrt{2}$ 1.5 $1\frac{5}{10}$ $\sqrt{3}$ 1.41 $1\frac{1}{3}$

Two of these are very close together. Which two, and which is larger? _____

Task 2.5 • True or false?

JUDGE

#	STATEMENT	T / F	REASON
a	Every fraction is a rational number.		
b	Every decimal that goes on forever is irrational.		
c	$\sqrt{16}$ is irrational.		
d	$22/7$ is exactly equal to π .		

Errors and one famous argument

STRETCH

17 MIN

Task 2.6 • Find the error

DETECTIVE

Dee

"0.333... goes on forever, so it is irrational"

What went wrong?

Correct answer: _____

Eli

" $\sqrt{25}$ is irrational because it has a root sign"

What went wrong?

Correct answer: _____

Fay

" $\pi = 3.14$ exactly"

What went wrong?

Correct answer: _____

Task 2.7 • Think it through

CHALLENGE

a. Let $x = 0.999\ldots$. Then $10x = 9.999\ldots$. Subtract the first from the second and solve for x . What do you get, and do you believe it?

b. Write a number that is between 0.3 and $\frac{1}{3}$. _____

c. Is $\sqrt{2} \times \sqrt{2}$ rational or irrational? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Never Ending

ANSWERS

Task 2.1 • Fraction to decimal

- a. $1/4 = 0.25$ e. $5/8 = 0.625$
 b. $1/3 = 0.333...$ f. $1/6 = 0.1666...$
 c. $3/8 = 0.375$ g. $7/20 = 0.35$
 d. $2/9 = 0.222...$ h. $1/7 = 0.142857...$

h repeats in a block of six digits and then starts again — 142857 forever. It is still rational, because it is 1 over 7.

Task 2.3 • Decimal to fraction

- a. $0.5 = 1/2$ e. $0.6 = 3/5$
 b. $0.25 = 1/4$ f. $0.125 = 1/8$
 c. $0.75 = 3/4$ g. $0.04 = 1/25$
 d. $0.2 = 1/5$ h. $1.5 = 3/2$

Every one starts as tenths, hundredths or thousandths and then cancels: 0.04 is 4/100, which is 1/25.

Task 2.2 • Sorting them

STOPS	0.5	$\sqrt{9}$ (=3)	0.7	-4	$2\frac{1}{2}$	$\sqrt{16}$ (=4)
REPEATS	$\frac{1}{3}$	0.161616...	22/7	(=3.142857...)		
NEITHER	$\sqrt{2}$	π	1.010010001...			

The third column holds the **irrational** numbers, and it is the only one that does. $\sqrt{9}$ and $\sqrt{16}$ are the traps — a root sign does not make a number irrational, and both of these are ordinary whole numbers in disguise. 22/7 is the other trap: it is a famous approximation of π but it is a fraction, so it repeats and it is rational. The last number is deliberately built to never repeat: one extra zero each time, forever.

Task 2.4 • Ordering

- 1.333... $1\frac{1}{3}$
 1.41 1.41
 1.41421 $\sqrt{2}$
 1.5 1.5
 1.5 $1\frac{5}{10}$
 1.732 $\sqrt{3}$

$1\frac{1}{3}$, 1.41 , $\sqrt{2}$, $1.5 = 1\frac{5}{10}$, $\sqrt{3}$

1.41 and $\sqrt{2}$ are the close pair, and $\sqrt{2}$ is the larger — 1.41421... against 1.41. Students who assume they are equal have made exactly Cal's mistake from Lesson 01. Note that 1.5 and $1\frac{5}{10}$ are the same number written two ways.

Task 2.5 • True or false?

- a. **true** that is the definition
 b. **false** 0.333... repeats, and is $\frac{1}{3}$
 c. **false** $\sqrt{16} = 4$
 d. **false** $22/7 = 3.142857...$
 $\pi = 3.141592...$

d matters. 22/7 is a very good approximation and it is not π . They part company at the third decimal place.

Task 2.7 • Challenge

- a. $x = 0.999...$
 $10x = 9.999...$
 $10x - x = 9.999... - 0.999...$
 $9x = 9$
 $x = 1$

a. The algebra is airtight and the conclusion is that **0.999... is exactly 1** — not nearly, not approaching, but equal. Expect disbelief; that is the correct response and the argument survives it. If a class pushes back, ask what number could possibly sit between the two.

b. Anything from 0.31 to 0.333, since $\frac{1}{3}$ is 0.3333... Accept 0.32, 0.33, or 0.3331.

c. **Rational — it is exactly 2.** Two irrational numbers multiplied together can land on a perfectly ordinary whole number, which is a genuinely surprising and true fact.

Money Over Time

READ BEFORE THE BELL

2 MINUTES

Students calculate simple interest, then compound interest year by year, and see the gap between the two open up. It finishes with a credit card at 20 per cent and a savings account at 4 per cent, and the same arithmetic running in both directions.

You do not need to know this subject

Simple interest is one multiplication. Compound interest is the same multiplication repeated, with the new total each time. Both formulas are on the student page and the key shows every year.

Photocopy these pages only

Pages 17–19

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A calculator is close to essential for this one.

WHAT TO SAY TO START

"Put a thousand dollars in an account paying five per cent. After twenty years, does it become two thousand or two thousand six hundred? The difference is the single most useful idea in personal finance, and it is just multiplication done repeatedly."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the interest panel	Simple uses the original amount every time
14–26	Task 3.1 — simple interest	One multiplication each
26–38	Task 3.2 — compounding year by year	Fill the table down
38–50	Task 3.3 — the credit card	Interest working against you
50–60	Challenge, collect sheets	Discussion question below

Questions students will ask

"What is the difference between the two?"

Simple interest is always worked out on the amount you started with. Compound interest is worked out on whatever you have now, including interest already earned.

"Do I have to do compound interest one year at a time?"

On this sheet, yes, and that is deliberate — doing it by hand is what makes the pattern visible. The shortcut formula is given in the challenge for anyone who wants it.

"Why does the credit card grow so fast?"

Same mechanism, pointed the other way. At 20 per cent a debt left alone roughly doubles in under four years, and Task 3.3 shows it happening.

If they finish early

Task 3.6 gives the doubling shortcut — divide 72 by the interest rate — and asks students to test it against their own table. It is startlingly accurate.

Curriculum focus • for the regular teacher's records

Grade 7–8: solving multi-step percentage problems including simple interest, and using repeated percentage change to model growth. Discussion question for the last five minutes: *"Why does money left alone grow faster and faster rather than steadily?"*

Two ways to earn interest

UNDERSTAND

20 MIN

NAME _____

CLASS _____

DATE _____

The same percentage, applied two different ways

SIMPLE INTEREST

$$I = P \times r \times t$$

P = amount you started with

r = rate as a decimal

t = number of years

\$1,000 at 5% for 3 years

$$I = 1000 \times 0.05 \times 3$$

$$= \$150$$

COMPOUND INTEREST

Work out one year, add it on, then start again with the new total.

year 1 1000 → **1050**year 2 1050 → **1102.50**year 3 1102.50 → **1157.63**interest = **\$157.63**

THE SHORTCUT

5% more means $\times 1.05$

$$1000 \times 1.05 = 1050$$

$$1050 \times 1.05 = 1102.50$$

10% more → $\times 1.10$ 20% more → $\times 1.20$ 3% more → $\times 1.03$

Task 3.1 • Simple interest

DO

#	AMOUNT	RATE	YEARS	INTEREST	TOTAL
a	\$1,000	5%	3		
b	\$500	4%	2		
c	\$2,400	6%	5		
d	\$8,000	3.5%	4		

Task 3.2 • Now compound it

DO

\$1,000 is invested at **5% compound**. Fill the table one year at a time, then compare the last column with simple interest.

YEAR	START WITH	INTEREST THIS YEAR	END WITH	SIMPLE WOULD GIVE
1	\$1,000.00			\$1,050.00
2				\$1,100.00
3				\$1,150.00
4				\$1,200.00
5				\$1,250.00

How much more does compounding give you after five years? _____

Debts and decisions

KEY SKILL

22 MIN

Task 3.3 • The credit card

APPLY

A debt of **\$2,000** sits on a card charging **20% a year**. Nothing is repaid.

YEAR	DEBT AT THE END
0	\$2,000.00
1	
2	
3	
4	

a. After how many years has the debt doubled? _____

b. How much of the year-4 debt is interest? _____

c. Simple interest at 20% for 4 years would add how much? _____

d. Which is worse for the borrower, and by how much? _____

Task 3.4 • Which account?

COMPARE

Account A pays **6% simple**. **Account B** pays **5% compound**. You have **\$4,000** to invest.

a. Value of A after 3 years? _____

b. Value of B after 3 years? _____

c. Which is ahead at 3 years? _____

d. Predict which is ahead at 20 years, and say why. _____

Doubling time

18 MIN

Task 3.5 • Find the error

DETECTIVE

Gus

\$1,000 at 5%
for 3 years compound
= $1000 \times 0.05 \times 3$
= \$150

What went wrong?

Correct answer: _____

Hana

"20% off then 20%
on gets you back
to where you
started"

What went wrong?

Correct answer: _____

Ira

"5% compound for
2 years is the
same as 10%
for 1 year"

What went wrong?

Correct answer: _____

Task 3.6 • Think it through

CHALLENGE

a. Bankers use a shortcut: **divide 72 by the interest rate** to find how many years money takes to double. Test it on your credit card table from Task 3.3 — $72 \div 20$ is 3.6 years. Does it match?

b. Use the shortcut: at 6%, how long to double? At 9%? _____

c. Why does compound interest beat simple interest eventually, even at a lower rate? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Money Over Time

ANSWERS

Task 3.1 • Simple interest

- a. $1000 \times 0.05 \times 3 = \150 total **\$1,150**
b. $500 \times 0.04 \times 2 = \40 total **\$540**
c. $2400 \times 0.06 \times 5 = \720 total **\$3,120**
d. $8000 \times 0.035 \times 4 = \$1,120$ total **\$9,120**

Task 3.2 • Compounding at 5%

yr	start	interest	end
1	1,000.00	50.00	1,050.00
2	1,050.00	52.50	1,102.50
3	1,102.50	55.13	1,157.63
4	1,157.63	57.88	1,215.51
5	1,215.51	60.78	1,276.28

Simple interest would give \$1,250, so compounding is **\$26.28 ahead** after five years. Small — and that is worth saying, because the whole force of compounding is in what happens over decades, not over five years. Accept a cent either way from rounding.

Task 3.3 • The credit card at 20%

0	\$2,000.00
1	\$2,400.00
2	\$2,880.00
3	\$3,456.00
4	\$4,147.20

- a. between years **3 and 4**
b. $4,147.20 - 2,000 = \$2,147.20$
c. simple: $2000 \times 0.2 \times 4 = \$1,600$

d. Compound is worse for the borrower by **\$547.20** over four years. Same rate, same debt, and the only difference is that the interest itself started earning interest.

Task 3.4 • Which account?

- a. A: $4000 \times 0.06 \times 3 = 720$
→ **\$4,720**
b. B: 4000×1.05^3
→ **\$4,630.50**
c. **A is ahead at 3 years**

d. **B wins over twenty years**, and comfortably: A reaches \$8,800 while B reaches about \$10,613. A adds the same \$240 every single year forever; B adds a bigger amount each year because it is charging 5% of a larger number each time. A higher simple rate wins early and always loses eventually.

Task 3.5 • The three errors

Gus — \$157.63. He used the simple interest formula for a compound question. Compounding has to be done year by year, or with 1.05^3 .

Hana — no. \$100 down 20% is \$80; \$80 up 20% is \$96. The second percentage is taken from a smaller number, so you never get back.

Ira — no, it is slightly more. $1.05 \times 1.05 = 1.1025$, so two years at 5% is 10.25%, not 10%. The extra quarter of a per cent is the compounding, and over decades that sliver is the whole story.

Task 3.6 • Challenge

a. **It matches almost exactly.** $72 \div 20 = 3.6$ years, and the table shows the debt passing \$4,000 between years 3 and 4. The rule of 72 is an approximation that has no business being as good as it is.

b. At 6%: $72 \div 6 = 12$ years. At 9%: $72 \div 9 = 8$ years. Worth noting that halving the rate more than doubles the wait.

c. Because simple interest always earns on the original amount, so it adds the same amount every year forever. Compound interest earns on everything you have, so each year it is taking a percentage of a bigger number. The simple account adds a fixed step; the compound account adds a growing one, and a growing step overtakes a fixed one sooner or later whatever the rates are.

Same Shape, Different Size

READ BEFORE THE BELL

2 MINUTES

Students find the scale factor between similar figures, use it to find missing sides, and then measure things too tall to reach — a tree, a building, a flagpole — using nothing but a shadow and a proportion. It ends with what happens to area when everything gets bigger.

You do not need to know this subject

Two shapes are similar if one is an enlargement of the other. Divide a pair of matching sides to get the scale factor, then multiply. Every step is in the key.

Photocopy these pages only

Pages 22–24

One per student. Do not photocopy this page or the answer key.

What students need

A pencil and a ruler.

WHAT TO SAY TO START

"You cannot climb a tree with a tape measure. But if you stand next to it on a sunny day, your shadow and the tree's shadow make two triangles that are exactly the same shape — and that is enough to work out how tall it is without leaving the ground."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the similarity panel	Matching sides, matching order
14–26	Task 4.1 — scale factors	Divide the pair
26–38	Task 4.2 — missing sides	Multiply or divide
38–50	Task 4.3 — shadows and mirrors	The best part
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"How do I know which sides match?"

The shapes are labelled in matching order, so the first side of one goes with the first side of the other. If a shape has been turned, look at the angles.

"Do I multiply or divide?"

Going from the small shape to the big one, multiply by the scale factor. Coming back the other way, divide.

"Does the area go up by the same factor?"

No, and Task 4.6 is entirely about this. If the sides triple, the area goes up nine times, because area needs two lengths.

If they finish early

Task 4.6 asks students to check whether two given rectangles are actually similar. One pair is and one pair is not, and the only way to tell is to compare both ratios.

Curriculum focus • for the regular teacher's records

Grade 7–8: solving problems involving scale drawings and similar figures, including finding missing lengths using proportional reasoning. Discussion question for the last five minutes: *"Why does making something twice as tall make it four times as heavy?"*

Scaling up and down

18 MIN

NAME _____

CLASS _____

DATE _____

Same shape means every side scales by the same number

SIMILAR

same angles
sides in the same ratio

small 3 4 5
big 6 8 10

$$6 \div 3 = 8 \div 4 = 10 \div 5 = 2$$

scale factor $k = 2$

FINDING A MISSING SIDE

small 4 ?
big 12 15

$$k = 12 \div 4 = 3$$

$$? = 15 \div 3 = 5$$

small $\rightarrow$ big $\times k$
big $\rightarrow$ small $\div k$

THE SHADOW TRICK

You 1.5 m tall
your shadow 2 m
tree shadow 18 m

$$18 \div 2 = 9$$

$$\text{tree} = 1.5 \times 9 = 13.5 \text{ m}$$

Task 4.1 • Find the scale factor

DO

#	SMALL SHAPE	BIG SHAPE	SCALE FACTOR	SIMILAR?
a	3, 4, 5	6, 8, 10		
b	2, 5	6, 15		
c	4, 6	6, 9		
d	3, 4	6, 9		
e	5, 12, 13	15, 36, 39		

One row is not similar at all. Which one, and how can you tell? _____

Task 4.2 • Missing sides

DO

These pairs are all similar. Find each missing length.

#	SMALL	BIG	MISSING	#	SMALL	BIG	MISSING
a	4, ?	12, 15		d	7, 3	?, 12	
b	5, 8	20, ?		e	6, 10	9, ?	
c	?, 9	10, 15		f	2.5, 4	10, ?	

Shadows, mirrors and ladders

KEY SKILL

20 MIN

Task 4.3 • Measuring what you cannot reach

APPLY

a. A person 1.5 m tall casts a 2 m shadow. A tree beside them casts an 18 m shadow. How tall is the tree?

b. A 1.8 m person casts a 3 m shadow. A building casts a 45 m shadow. How tall is the building?

c. A flagpole casts a 12 m shadow at the same time a 1.6 m student casts a 2.4 m shadow. How tall is the flagpole?

d. At noon the sun is overhead and shadows are almost nothing. Why does the method stop working?

Task 4.4 • A ladder against a wall

APPLY

A ladder leans against a wall. A 1 m stick placed under it touches the ladder at a height of **0.4 m**, standing **1 m** from the base. The ladder's foot is **3 m** from the wall.

a. Scale factor from the small triangle to the big one?

c. Why are the two triangles similar?

b. How high up the wall does the ladder reach? _____

d. If the foot moved to 4.5 m out, how high would it reach?

Ratios and areas

17 MIN

Task 4.5 • Find the error

DETECTIVE

Jo

small 3, 4
big 6, 7
“add 3 to each side”

What went wrong?

Correct answer: _____

Kit

“The sides tripled
so the area
tripled too”

What went wrong?

Correct answer: _____

Lex

small 4, 6
big 6, 9
“not similar,
the numbers are
different”

What went wrong?

Correct answer: _____

Task 4.6 • Think it through

CHALLENGE

- a. Is a $4\text{ cm} \times 6\text{ cm}$ rectangle similar to a $6\text{ cm} \times 9\text{ cm}$ one? Show both ratios. _____
- b. Is a $3\text{ cm} \times 4\text{ cm}$ rectangle similar to a $6\text{ cm} \times 9\text{ cm}$ one? _____
- c. A square has sides of 5 cm. Its sides are tripled. What happens to its perimeter, and what happens to its area? Work out both.

- d. Are all squares similar to each other? Are all rectangles? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Same Shape, Different Size

ANSWERS

Task 4.1 • Scale factors

- a. $6 \div 3 = 8 \div 4 = 10 \div 5 = 2$ **similar**
b. $6 \div 2 = 15 \div 5 = 3$ **similar**
c. $6 \div 4 = 9 \div 6 = 1.5$ **similar**
d. $6 \div 3 = 2$ but $9 \div 4 = 2.25$ **NOT similar**
e. $15 \div 5 = 36 \div 12 = 39 \div 13 = 3$ **similar**

d is the odd one out. The only way to tell is to check every pair — one matching ratio proves nothing. A student who checked only the first pair would have called it similar.

Task 4.2 • Missing sides

- a. $k = 3$? = $15 \div 3 = 5$
b. $k = 4$? = $8 \times 4 = 32$
c. $k = 10 \div ? \dots 15 \div 9 = 5/3$
? = $10 \div (5/3) = 6$
d. $k = 12 \div 3 = 4$? = $7 \times 4 = 28$
e. $k = 9 \div 6 = 1.5$? = **15**
f. $k = 10 \div 2.5 = 4$? = **16**

In c and d the given pair is the second one, so find k from those two and then work on the first. Order matters less than consistency.

Task 4.3 • Measuring what you cannot reach

- a. $18 \div 2 = 9 \rightarrow 1.5 \times 9 = 13.5$ m
b. $45 \div 3 = 15 \rightarrow 1.8 \times 15 = 27$ m
c. $12 \div 2.4 = 5 \rightarrow 1.6 \times 5 = 8$ m

d. With the sun directly overhead both shadows shrink towards nothing, and dividing one tiny measurement by another makes any small error enormous. The method needs a decent shadow, which is why surveyors do this in the morning or late afternoon. Any answer noticing that the numbers become unreliable is full marks.

Task 4.4 • The ladder

small triangle 1 m out, 0.4 m up
big triangle 3 m out, ? up

- a. $k = 3 \div 1 = 3$
b. $0.4 \times 3 = 1.2$ m up the wall
c. $k = 4.5$, so $0.4 \times 4.5 = 1.8$ m

c. Both triangles share the angle at the ladder's foot and both have a right angle at the ground, so their third angles must match too. Same angles means same shape means similar.

Task 4.5 • The three errors

Jo — similar shapes multiply, they do not add. 3, 4 scaled by 2 gives 6, 8. Her 6, 7 has changed the shape.

Kit — the area went up nine times. Area needs two lengths, and both of them tripled: $3 \times 3 = 9$.

Lex — they are similar. $6 \div 4$ and $9 \div 6$ both give 1.5. Different numbers is exactly what similar shapes have; it is the ratios that must match.

Task 4.6 • Challenge

- a. $6 \div 4 = 1.5$ $9 \div 6 = 1.5 \rightarrow$ **similar**
b. $6 \div 3 = 2$ $9 \div 4 = 2.25 \rightarrow$ **not similar**
c. 5 cm square: P = 20 cm A = 25 cm²
15 cm square: P = 60 cm A = 225 cm²
perimeter $\times 3$, area $\times 9$

c is the whole idea of the lesson in one line. Perimeter is a length so it scales by k. Area is two lengths so it scales by k². That is why a model car at one-tenth scale needs a hundredth of the paint.

d. All squares are similar; rectangles are not. Every square has the same shape, only the size changes. A 1 $\times$ 2 rectangle and a 1 $\times$ 5 rectangle are genuinely different shapes.

In and Out

2 MINUTES

Students complete input-output tables, work out the rule behind one, decide whether a relationship is a function at all, and meet $f(x)$ notation for the first time. It closes on a real cost formula and the question of which input makes sense.

You do not need to know this subject

A function is a machine: one input, one output, every time. The tables do the work and the key gives every rule and every value.

Photocopy these pages only

Pages 27–29

One per student. Do not photocopy this page or the answer key.

What students need

A pencil.

WHAT TO SAY TO START

“A vending machine that gave you a different snack every time you pressed the same button would be broken. That is exactly what a function is not allowed to do, and that one rule is the whole definition.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the machine panel	One input, one output
14–26	Task 5.1 — fill the tables	Apply the rule
26–36	Task 5.2 — find the rule	Work backwards
36–46	Task 5.3 — is it a function?	The definition, tested
46–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“What makes something a function?”

Each input gives exactly one output. An input that produces two different outputs breaks it. Two inputs sharing an output is fine.

“What does $f(3)$ mean?”

Put 3 in and see what comes out. It does not mean f multiplied by 3, and that misreading is the one to head off early.

“How do I find the rule from a table?”

Look at what happens to the output when the input goes up by one. That gives the multiplier. Then check what you need to add to make the first row work.

If they finish early

Task 5.6 gives a cost rule and asks which inputs make sense in real life. Negative and fractional inputs are mathematically fine and physically absurd, and noticing that is the point.

Curriculum focus • for the regular teacher’s records

Grade 8: understanding a function as a rule assigning exactly one output to each input, comparing functions represented in tables, rules and words.

Discussion question for the last five minutes: *“Can two different inputs give the same output and still be a function?”*

Machines and rules

UNDERSTAND

18 MIN

NAME _____

CLASS _____

DATE _____

A machine with one rule

THE RULE: $y = 3x + 2$

in	→	out
0	→	2
1	→	5
2	→	8
3	→	11

same input, same output,
every time

 $f(x)$ NOTATION

$$f(x) = 3x + 2$$

$$f(0) = 2$$

$$f(1) = 5$$

$$f(4) = 14$$

$f(4)$ means “put 4 in”.
It does **not** mean $f \times 4$.

NOT A FUNCTION

in	→	out
1	→	5
2	→	7
1	→	9 ← broken

1 gave two different
answers. One input, one
output – or it is
not a function.

Task 5.1 • Run the machine

DO

a. $y = 2x + 1$

IN	OUT
0	
1	
2	
5	
10	

b. $y = 4x - 3$

IN	OUT
0	
1	
2	
5	
10	

c. $f(x) = x^2$

IN	OUT
0	
1	
2	
5	
-3	

Task 5.2 • Find the rule

REVERSE

Each table follows a rule of the form $y = ?x + ?$. Find it.

#	INPUT	OUTPUT	RULE	OUTPUT AT $x = 10$
a	1, 2, 3, 4	5, 8, 11, 14		
b	0, 1, 2, 3	7, 9, 11, 13		
c	1, 2, 3, 4	2, 4, 6, 8		
d	0, 1, 2, 3	10, 7, 4, 1		

Testing the definition

KEY SKILL

20 MIN

Task 5.3 • Is it a function?

JUDGE

Write yes or no and give the reason.

#	THE RELATIONSHIP	FUNCTION?	WHY
a	inputs 1, 2, 3 → outputs 4, 5, 6		
b	inputs 1, 2, 1 → outputs 4, 5, 9		
c	inputs 1, 2, 3 → outputs 7, 7, 7		
d	each student → their date of birth		
e	each date of birth → students born that day		

Task 5.4 • A real function

APPLY

A printing shop charges a \$5 setup fee plus \$3 for each poster. The cost function is $C(n) = 3n + 5$.a. $C(0) =$ _____ what does that mean?

b. $C(10) =$ _____c. $C(25) =$ _____

d. You have \$50. How many posters? _____

e. Does $C(-4)$ make sense? _____f. Does $C(2.5)$? _____

Edges of the definition

STRETCH

17 MIN

Task 5.5 • Find the error

DETECTIVE

Mo

$$f(x) = 3x + 2$$

$$“f(4) = f \times 4”$$

What went wrong?

Correct answer: _____

Nel

“Inputs 1, 2, 3 all
giving output 7 is
not a function”

What went wrong?

Correct answer: _____

Oz

in 1 2 3 4
out 5 8 11 14
“the rule is
 $y = x + 3$ ”

What went wrong?

Correct answer: _____

Task 5.6 • Think it through

CHALLENGE

- a. Can two different inputs give the same output and still be a function? Give an example. _____
- b. For $f(x) = x^2$, find two inputs that both give 25. _____
- c. Write a rule where the output goes *down* as the input goes up. _____
- d. $f(x) = 2x + b$ and $f(3) = 11$. Find b . _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

In and Out

ANSWERS

Task 5.1 • Run the machine

a. $y = 2x + 1$ $= x^2$	b. $y = 4x - 3$	c. $f(x)$
$0 \rightarrow 1$	$0 \rightarrow -3$	$0 \rightarrow 0$
$1 \rightarrow 3$	$1 \rightarrow 1$	$1 \rightarrow 1$
$2 \rightarrow 5$	$2 \rightarrow 5$	$2 \rightarrow 4$
$5 \rightarrow 11$	$5 \rightarrow 17$	$5 \rightarrow 25$
$10 \rightarrow 21$	$10 \rightarrow 37$	$-3 \rightarrow 9$

The last row of c is the interesting one: a negative input gives a positive output, because squaring kills the sign.

Task 5.2 • Find the rule

- a. up 3 each time $\rightarrow y = 3x + 2$ $x=10 \rightarrow 32$
 b. up 2 each time $\rightarrow y = 2x + 7$ $x=10 \rightarrow 27$
 c. up 2 each time $\rightarrow y = 2x$ $x=10 \rightarrow 20$
 d. down 3 each time $\rightarrow y = 10 - 3x$ $x=10 \rightarrow -20$

The step between outputs is always the number in front of x. d steps down, so the multiplier is negative.

Task 5.3 • Is it a function?

- a. **yes** every input has one output
 b. **no** input 1 gave both 4 and 9
 c. **yes** repeated outputs are fine
 d. **yes** a person has exactly one birthday
 e. **no** one date can have several students

c and e are the pair that teaches the definition. Outputs are allowed to repeat as much as they like — that is c. What is forbidden is one *input* having two outputs, which is exactly what e does. Reversing a function does not always give a function back, and d and e are the same information read in opposite directions.

Task 5.4 • A real function

- a. $C(0) = \$5$ the setup fee with no posters
 b. $C(10) = \$35$
 c. $C(25) = \$80$
 d. $3n + 5 = 50 \rightarrow n = 15 \rightarrow 15$ posters

e and f are the point of the task. The rule happily computes $C(-4) = -\$7$ and $C(2.5) = \$12.50$, but you cannot print minus four posters, and half a poster is not a thing the shop sells. The formula has a wider domain than the situation does, and knowing where the maths stops describing reality is a real skill.

Task 5.5 • The three errors

Mo — $f(4) = 14$. The brackets mean “put 4 in for x”, not multiply. This misreading is worth stamping out on the spot.

Nel — it is a function. Every input has exactly one output. Nothing says the outputs have to be different.

Oz — $y = 3x + 2$. His rule works for the first row and nothing else: $x = 2$ would give 5, not 8. The outputs go up in threes, not ones.

Task 5.6 • Challenge

- a. **Yes.** $y = x^2$ sends both 3 and -3 to 9. Also acceptable: a constant rule such as $y = 7$, which sends everything to the same place and is still perfectly well behaved.
 b. **5 and -5.**
 c. Anything with a negative multiplier: $y = 10 - 2x$, or $y = -x$. Accept a real context too, such as fuel left in a tank against distance driven.
 d. **b = 5.** $f(3) = 6 + b = 11$.

Two Lines, One Answer

READ BEFORE THE BELL

2 MINUTES

Students solve pairs of linear equations by graphing them and reading off the crossing point, check the answer by substituting into both, and then meet the two cases where lines never cross or lie on top of each other. It finishes with two hire companies and the day one becomes cheaper.

You do not need to know this subject

Everything is done on the printed grids and every crossing point lands on a whole-number pair. Students only need to plot two points per line and join them with a ruler.

Photocopy these pages only**Pages 32–34**

One per student. Do not photocopy this page or the answer key.

What students need

A pencil and a ruler. Coloured pencils help but are not needed.

WHAT TO SAY TO START

“Two rules, two lines, one place where they meet. That crossing point is the only pair of numbers that satisfies both rules at once — which is exactly what you need when two hire companies are competing for your money.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the graphing panel	Two points make a line
14–28	Task 6.1 — plot and read off	Ruler, please
28–38	Task 6.2 — check by substituting	Both equations, every time
38–50	Task 6.3 — no crossing, or every point	The two odd cases
50–60	Challenge, collect sheets	Discussion question below

Questions students will ask**“How do I draw the line?”**

Pick two easy x values, work out y for each, plot the two points and join them with a ruler. A third point is a useful check.

“How do I know the crossing point is right?”

Put it into *both* equations. If both come out true, it is the answer. If only one does, the reading is wrong.

“What if the lines never meet?”

Then there is no solution, and the reason is always that the two lines have the same slope. Task 6.4 has one of each case.

If they finish early

Task 6.5 gives two hire companies and asks who is cheaper, and the honest answer is “it depends” — which is the whole reason for finding the crossing point.

Curriculum focus • for the regular teacher’s records

Grade 8: analysing and solving pairs of simultaneous linear equations, including by graphing, and interpreting the point of intersection. Discussion question for the last five minutes: “What does the crossing point of two lines actually mean?”

Where the lines meet

UNDERSTAND

16 MIN

NAME _____

CLASS _____

DATE _____

Two lines, and the one point they share

$$y = 2x + 1$$

x	0	1	2	3
y	1	3	5	7

Plot two of these and join them with a ruler.

$$y = -x + 7$$

x	0	2	4	6
y	7	5	3	1

They cross at **(2, 5)**.

ALWAYS CHECK

put (2, 5) into both:

$$5 = 2(2) + 1 = 5 \checkmark$$

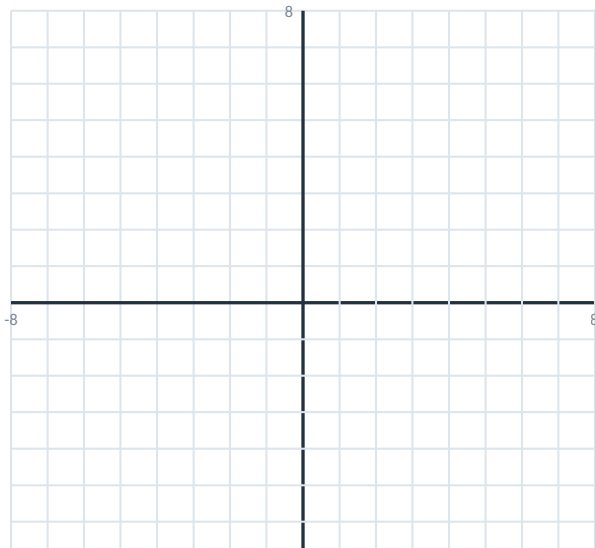
$$5 = -(2) + 7 = 5 \checkmark$$

Both true, so it is the answer.

Task 6.1 • Plot both and read it off

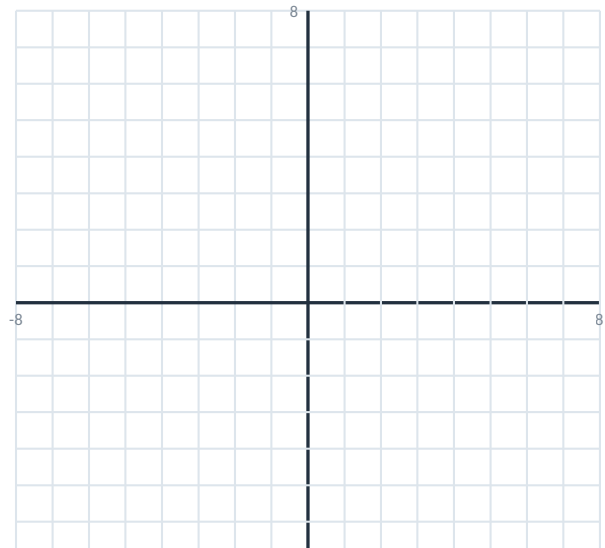
PLOT

a. $y = 2x + 1$ and $y = -x + 7$



They cross at _____

b. $y = x + 2$ and $y = -2x + 8$



They cross at _____

Checking, parallels and real hire

KEY SKILL

26 MIN

Task 6.2 • Check by substituting

VERIFY

Somebody claims each pair below is the crossing point. Substitute into **both** equations and say whether they are right.

#	THE TWO EQUATIONS	CLAIMED POINT	FIRST CHECK	SECOND CHECK	RIGHT?
a	$y = 3x - 1$ / $y = x + 5$	(3, 8)			
b	$y = 2x$ / $y = x + 4$	(4, 8)			
c	$y = x - 3$ / $y = -x + 5$	(2, -1)			

Task 6.3 • When they never meet

THINK

For each pair, work out the slope of both lines *without drawing them*, then say how many crossing points there are.

#	THE TWO EQUATIONS	SLOPES	ONE POINT / NONE / EVERY POINT
a	$y = 2x + 1$ and $y = 2x + 5$		
b	$y = 3x - 2$ and $y = -x + 6$		
c	$y = x + 4$ and $2y = 2x + 8$		
d	$y = 4x$ and $y = 4x - 7$		

What do the two “never meet” pairs have in common? _____

Task 6.4 • Two hire companies

APPLY

Rapid Hire charges \$60 plus \$20 a day. **Budget Vans** charges \$100 plus \$10 a day.

DAYS	RAPID	BUDGET
1		
2		
4		
6		
8		

- Rule for each company. _____
- On which day do they cost the same? _____
- What is the cost that day? _____
- Which one for a weekend? _____
- Which one for a fortnight? _____

Parallel, identical, impossible

STRETCH

16 MIN

Task 6.5 • Find the error

DETECTIVE

Pat

"I checked (3, 8) in the first equation and it worked, so it is the answer"

What went wrong?

Correct answer: _____

Quin

" $y = 2x + 1$ and $y = 2x + 5$ must cross somewhere further along"

What went wrong?

Correct answer: _____

Ros

"The lines cross at about (2.4, 5.7) so that is the answer"

What went wrong?

Correct answer: _____

Task 6.6 • Think it through

CHALLENGE

- Write a second equation so that it and $y = 3x + 1$ never cross. _____
- Write a second equation so that every point on $y = 3x + 1$ is a solution. _____
- Can two straight lines cross at exactly two points? Explain. _____
- A company with a higher joining fee but a lower daily rate is better for whom? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Two Lines, One Answer

ANSWERS

Task 6.1 • Plot and read off

a. $y = 2x + 1$ x 0 1 2 3
 y 1 3 5 7
 $y = -x + 7$ x 0 2 4 6
 y 7 5 3 1
 cross at **(2, 5)**

b. $y = x + 2$ x 0 2 4
 y 2 4 6
 $y = -2x + 8$ x 0 2 4
 y 8 4 0
 cross at **(2, 4)**

Task 6.2 • Check by substituting

a. (3, 8) $3(3) - 1 = 8$ ✓ $3 + 5 = 8$ ✓ **right**
 b. (4, 8) $2(4) = 8$ ✓ $4 + 4 = 8$ ✓ **right**
 c. (2, -1) $2 - 3 = -1$ ✓ $-2 + 5 = 3$ ✗ **wrong**

c passes the first test and fails the second, which is exactly why both have to be checked. The real crossing point for c is (4, 1).

Task 6.3 • When they never meet

- a. slopes 2 and 2, different intercepts → **none** (parallel)
- b. slopes 3 and -1 → **one point**
- c. $2y = 2x + 8$ is $y = x + 4$, the same line → **every point**
- d. slopes 4 and 4, different intercepts → **none** (parallel)

a and d share the same slope in both equations. Equal slopes means the lines rise at exactly the same rate, so the gap between them never closes. c is the trap: it looks like a different equation until you divide through by 2 and find it is the same line written twice.

Task 6.4 • Two hire companies

days	1	2	4	6	8
Rapid	80	100	140	180	220
Budget	110	120	140	160	180

- a. $R = 20d + 60$ $B = 10d + 100$
- b. $20d + 60 = 10d + 100$
 $10d = 40$ → **day 4**
- c. **\$140** either way
- d. weekend (2 days) → **Rapid**
- e. fortnight → **Budget**

Neither company is cheaper in general, and that is the honest answer to give a class that wants one. The crossing point is the whole information: below four days Rapid wins, above four days Budget does.

Task 6.5 • The three errors

Pat — one check is not enough. A point on one line is not a solution unless it is on the other as well. Both equations, every time.

Quin — they never cross. Both climb 2 units for every 1 across, so the vertical gap of 4 stays 4 forever, however far you extend them.

Ros — read it again. On these grids every crossing point sits exactly on a corner. A reading like (2.4, 5.7) means the lines were drawn by eye rather than through plotted points.

Task 6.6 • Challenge

- a. Any line with slope 3 and a different intercept: $y = 3x + 5$, or $y = 3x - 2$. Same slope, different starting point, never meet.
- b. Any disguised version of the same line: $2y = 6x + 2$, or $y - 1 = 3x$. Every point on one is a point on the other.
- c. **No.** Two straight lines that share two points are the same line, and then they share all of them. So the only possibilities are none, one, or infinitely many.
- d. Better for the heavy user. A high fixed cost is spread across more days, and the lower daily rate then wins — which is precisely what happened to Budget Vans after day four.

Many Sides

2 MINUTES

Students find the interior angle sum of any polygon using the triangle rule, work out the size of one angle in a regular polygon, and discover that the exterior angles of every polygon ever drawn add to exactly 360° no matter how many sides it has.

You do not need to know this subject

One formula, $(n - 2) \times 180$, and it is on the student page along with a worked example. The rest is dividing.

Photocopy these pages only

Pages 37–39

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A ruler is useful for the drawing task.

WHAT TO SAY TO START

“A triangle’s angles add to 180° . Everybody knows that. What almost nobody knows is that it tells you the answer for a shape with a hundred sides too, and today you work out how.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the triangle-splitting panel	Count the triangles
14–26	Task 7.1 — angle sums	Straight substitution
26–36	Task 7.2 — regular polygons	Divide by n
36–46	Task 7.3 — exterior angles	The surprise
46–50	Challenge, collect sheets	Discussion question below

Questions students will ask

“Where does the formula come from?”

Split the shape into triangles from one corner. A quadrilateral makes 2, a pentagon 3, a hexagon 4 — always two fewer than the number of sides. Each triangle brings 180° .

“What is an exterior angle?”

The angle between one side and the extension of the next. Interior and exterior at the same corner always add to 180° .

“Why do exterior angles always total 360° ?”

Because walking all the way round the shape turns you through one complete circle. Task 7.5 asks students to picture exactly that.

If they finish early

Task 7.5 asks how many sides a regular polygon has if each interior angle is 150° . Working backwards through the formula is a genuine step up.

Curriculum focus • for the regular teacher’s records

Grade 8: establishing facts about the angle sum of polygons and the exterior angles of polygons, and using them to solve problems. Discussion question for the last five minutes: “Why does a shape with a thousand sides still have exterior angles adding to only 360° ?”

Counting the triangles

UNDERSTAND

13 MIN

NAME _____

CLASS _____

DATE _____

Every polygon is triangles in disguise

SPLIT IT FROM ONE CORNER

triangle	1
triangle	180°
quadrilateral	
2	360°
pentagon	
3	540°
hexagon	
4	720°
octagon	6
1080°	

Always $n - 2$
triangles.

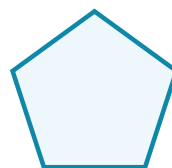
THE FORMULA

$$\text{sum} = (n - 2) \times 180$$

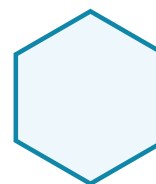
$$n = 5 \quad (5 - 2) \times 180 = \mathbf{540^\circ}$$

REGULAR means all
angles
equal, so divide by n :

$$540 \div 5 = \mathbf{108^\circ} \text{ each}$$



pentagon 540°



hexagon 720°

Task 7.1 • Interior angle sums

DO

#	SHAPE	SIDES (N)	TRIANGLES	ANGLE SUM
a	triangle	3		
b	quadrilateral	4		
c	pentagon	5		
d	hexagon	6		
e	octagon	8		
f	decagon	10		

Regular shapes and outside angles

KEY SKILL

23 MIN

Task 7.2 • One angle in a regular polygon

DO

#	REGULAR SHAPE	ANGLE SUM	÷ N	EACH INTERIOR ANGLE
a	equilateral triangle			
b	square			
c	regular pentagon			
d	regular hexagon			
e	regular octagon			

Task 7.3 • Exterior angles

DISCOVER

At every corner the interior and exterior angles sit on a straight line, so they add to 180° . Use your answers from Task 7.2.

#	REGULAR SHAPE	INTERIOR ANGLE	EXTERIOR ANGLE	× NUMBER OF SIDES
a	equilateral triangle			
b	square			
c	regular pentagon			
d	regular hexagon			
e	regular octagon			

Look down your last column. What do you notice, and does it depend on the number of sides at all?

Task 7.4 • Find the missing angle

APPLY

a. A quadrilateral has angles 85° , 100° and 62° . Find the fourth.

b. A pentagon has four angles of 110° . Find the fifth.

c. A regular polygon has an exterior angle of 45° . How many sides?

d. A hexagon has five angles of 125° . Find the sixth.

Tiling and turning

STRETCH

14 MIN

Task 7.5 • Find the error

DETECTIVE

Sid

"A hexagon has 6 sides so its angles add to $6 \times 180 = 1080^\circ$ "

What went wrong?

Correct answer: _____

Tam

"A polygon with 20 sides has exterior angles adding to 7200° "

What went wrong?

Correct answer: _____

Uri

"Every pentagon has five 108° angles"

What went wrong?

Correct answer: _____

Task 7.6 • Think it through

CHALLENGE

- a. A regular polygon has interior angles of 150° . How many sides does it have? _____
- b. Why do regular hexagons tile a floor with no gaps, but regular pentagons cannot? _____
- c. Imagine walking right around the outside of any polygon. How far have you turned by the time you are back where you started? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Many Sides

ANSWERS

Task 7.1 • Interior angle sums

- a. $n=3$ 1 triangle **180°**
- b. $n=4$ 2 triangles **360°**
- c. $n=5$ 3 triangles **540°**
- d. $n=6$ 4 triangles **720°**
- e. $n=8$ 6 triangles **1080°**
- f. $n=10$ 8 triangles **1440°**

The triangle count is always two fewer than the sides, because the two sides next to the corner you split from cannot form triangles of their own.

Task 7.2 • Regular polygons

- a. $180 \div 3 = 60^\circ$
- b. $360 \div 4 = 90^\circ$
- c. $540 \div 5 = 108^\circ$
- d. $720 \div 6 = 120^\circ$
- e. $1080 \div 8 = 135^\circ$

Every one of these gets larger as the shape gains sides, creeping towards 180° without ever reaching it — which is why a many-sided polygon starts to look like a circle.

Task 7.3 • Exterior angles

shape	interior	exterior	$\times n$
triangle	60°	120°	360°
square	90°	90°	360°
pentagon	108°	72°	360°
hexagon	120°	60°	360°
octagon	135°	45°	360°

Every single row gives 360°, and it does not depend on the number of sides at all. This is the moment of the lesson. It is true for any polygon, regular or not, with three sides or three thousand — and the reason is in Task 7.6c.

Task 7.4 • Missing angles

- a. $360 - (85+100+62) = 113^\circ$
- b. $540 - (4 \times 110) = 100^\circ$
- c. $360 \div 45 = 8 \text{ sides}$
- d. $720 - (5 \times 125) = 95^\circ$

c uses the exterior-angle fact rather than the interior formula, and it is far quicker that way. A student who went the long way round via $(n - 2) \times 180$ still gets 8, and should be told so.

Task 7.5 • The three errors

Sid — 720°. The formula is $(n - 2) \times 180$, not $n \times 180$. His answer is the octagon's sum.

Tam — 360°. Exterior angles always total 360° whatever n is. He has used the interior rule by mistake.

Uri — only regular ones. Any pentagon's angles add to 540°, but they only split evenly into five 108° angles if the shape is regular. A long thin pentagon is still a pentagon.

Task 7.6 • Challenge

a. Twelve sides. If each interior angle is 150° then each exterior is 30°, and $360 \div 30 = 12$. Going through the exterior angle is far easier than solving $(n - 2) \times 180 = 150n$, though that works too.

b. Hexagons have 120° angles and three of them meet exactly at 360°. Pentagons have 108° angles: three make 324° and leave a gap, four make 432° and overlap. The angle has to divide into 360° exactly, and 108 does not.

c. One full turn — 360°. Each exterior angle is exactly the turn you make at that corner, and by the time you are home you are facing the way you started. That is the whole proof, and it works for any polygon at all.

Scattered

2 MINUTES

Students plot two-variable data, describe the trend as positive, negative or absent, draw a line of best fit by eye and use it to predict. It closes on the difference between two things moving together and one of them causing the other.

You do not need to know this subject

The grid is printed and the points are given as coordinate pairs. Drawing the line of best fit is done by eye with a ruler, so there is no calculation to get wrong.

Photocopy these pages only

Pages 42–44

One per student. Do not photocopy this page or the answer key.

What students need

A pencil and a ruler. A clear ruler is ideal for the line of best fit.

WHAT TO SAY TO START

“Children with bigger feet read better. That is a real, measured, reliable finding, and it is completely true. It also does not mean what it sounds like it means, and by the end of today you will know exactly what is going on.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the correlation panel	Three shapes to recognise
14–26	Task 8.1 — plot and describe	Ruler for the line
26–36	Task 8.2 — name the correlation	Quick and discussable
36–48	Task 8.3 — predicting from the line	Interpolation is safe, extrapolation is not
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Does the line have to touch the points?”

No. A line of best fit runs through the middle of the cloud with roughly as many points above it as below. It is not meant to hit any of them.

“What is the difference between strong and weak?”

How tightly the points hug the line. A tight band is strong; a scattered cloud with a faint slope is weak. Both can be positive.

“Does correlation prove cause?”

No, and Task 8.6 is entirely about this. Two things can move together because a third thing moves both of them.

If they finish early

Task 8.6 asks why shoe size and reading ability go together in primary-school children. The answer is age, and the moment a class sees it is the best thirty seconds in the lesson.

Curriculum focus • for the regular teacher’s records

Grade 8: constructing and interpreting scatter plots for bivariate data, describing patterns of association, and fitting a straight line informally to assess the fit. Discussion question for the last five minutes: *“If two things always happen together, how would you find out whether one causes the other?”*

Reading a cloud of points

UNDERSTAND

14 MIN

NAME _____

CLASS _____

DATE _____

Three shapes a cloud of points can make

POSITIVE

one goes up, so does
the other

up to the right ↗

hours studied and
test score

NEGATIVE

one goes up, the
other
goes down

down to the right ↘

age of a car and
what it is worth

NONE

no pattern at all

just a blob

shoe size and
house number

LINE OF BEST FIT

Straight line
through the
middle of the cloud.

Roughly as many
points
above as **below**.

It does not have to
touch
any of them.

Task 8.1 • Plot it and describe it

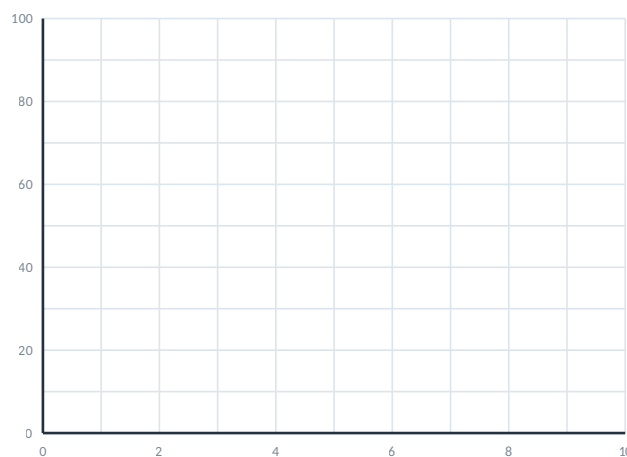
PLOT

Five students recorded how many hours they revised and what they scored.

hours	1	2	3	5	6
score	52	58	66	79	84

- Plot all five points. _____
- Is the correlation positive, negative or absent?

- Is it strong or weak? _____
- Draw a line of best fit with a ruler.



hours revised across, score up the side

Naming, predicting, pushing too far

KEY SKILL

24 MIN

Task 8.2 • Name the correlation

JUDGE

Write positive, negative or none.

#	THE TWO THINGS	CORRELATION
a	hours of revision and test score	
b	the age of a car and its value	
c	a student's height and their house number	

#	THE TWO THINGS	CORRELATION
d	outside temperature and hot chocolate sales	
e	a person's height and their shoe size	
f	distance run and time taken	

Task 8.3 • Predicting from the line

APPLY

Use the line of best fit you drew in Task 8.1.

- a. Predict the score for **4 hours**. _____
- b. Predict the score for **7 hours**. _____
- c. Predict the score for **0 hours**. _____

- d. Predict the score for **20 hours**. _____
- e. Which of your four predictions do you trust least, and why?

Task 8.4 • Strong, weak, or neither

JUDGE

Four classes plotted the same two variables and got four different clouds.

#	WHAT THE PLOT LOOKED LIKE	DIRECTION	STRENGTH
a	points almost exactly on a line going up		
b	a wide scattered band drifting up		
c	a tight band going down		
d	a round blob with no slope		

Which would you trust most for making a prediction? _____

Correlation is not cause

STRETCH

17 MIN

Task 8.5 • Find the error

DETECTIVE

Vi

"My line misses every point, so I drew it wrong"

What went wrong?

Correct answer: _____

Wes

"They are strongly correlated, so one must be causing the other"

What went wrong?

Correct answer: _____

Zoe

joins each point to the next one and calls the zigzag a line of best fit

What went wrong?

Correct answer: _____

Task 8.6 • Think it through

CHALLENGE

a. In primary schools, children with bigger feet read better. The correlation is strong and real. Suggest a third thing that explains both.

b. Your line predicts a score above 100 for enough hours of revision. Why is that a problem with the method rather than with the data?

c. Sketch, in words, two variables you would expect to show no correlation at all. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Scattered

ANSWERS

Task 8.1 • Plot it and describe it

points: (1, 52) (2, 58) (3, 66) (5, 79) (6, 84)

b. **positive** c. **strong** – the points sit very close to a straight line

A good line of best fit passes near (0, 46) and (6, 84), climbing about 6.5 marks per hour of revision. Any straight line through the middle of the cloud with a similar slope is correct — there is no single right answer, and telling a class that is worth doing explicitly, because they will assume there is one.

Task 8.2 • Name the correlation

- | | |
|--------------------------|-----------------|
| a. revision / score | positive |
| b. car age / value | negative |
| c. height / house number | none |
| d. temperature / cocoa | negative |
| e. height / shoe size | positive |
| f. distance / time | positive |

d catches people who read “temperature up” and reach for positive. Hot chocolate sells when it is cold, so the two move in opposite directions.

Task 8.4 • Strong, weak, or neither

- a. **positive, strong**
 b. **positive, weak**
 c. **negative, strong**
 d. **none** – strength does not apply

a and c are equally trustworthy for prediction, and that is the point to draw out: a strong negative relationship is exactly as useful as a strong positive one. Direction and strength are two separate questions.

Task 8.3 • Predicting from the line

- | | |
|--------------------------------|-------------------------------------|
| a. 4 hours → about 72 | inside the data – safe |
| b. 7 hours → about 91 | just outside – probably fine |
| c. 0 hours → about 46 | just outside – plausible |
| d. 20 hours → about 176 | far outside – nonsense |

Accept anything within two or three marks of these; the readings come off a hand-drawn line. **e is the answer that matters:** d. Predicting inside the range you measured is called interpolation and it is reasonable. Predicting far outside it is extrapolation, and it assumes the pattern keeps going when nothing in the data says it does. Nobody scores 176.

Task 8.5 • The three errors

Vi — that is what a line of best fit does. It runs through the middle of the cloud, not through the points. A line that hit every point would mean the relationship was perfect, which real data never is.

Wes — no. Strength tells you how tightly the two move together, and nothing at all about why. See the challenge.

Zoe — a line of best fit is straight. Joining the dots describes the five students she measured; a straight line describes the relationship, which is what lets her predict a sixth.

Task 8.6 • Challenge

a. Age. Older primary children have bigger feet, and older primary children read better. Feet and reading have nothing to do with each other — a third variable is driving both, and it produces a correlation that is entirely real and entirely meaningless. Accept “how old they are” or “year level”.

b. Because the line is a model of the range that was measured, and it knows nothing about the ceiling of 100. The data ran from 1 to 6 hours; the line has no authority beyond that. This is the same point as Task 8.3d.

c. Anything genuinely unrelated: shoe size and favourite colour, house number and test score, length of surname and running speed. If a student proposes a pair that is secretly linked, that is a good conversation rather than a wrong answer.

Chance Twice

2 MINUTES

Students list the sample space for two events happening one after another, draw tree diagrams, multiply probabilities along a branch, and see what changes when the first thing taken out is not put back. This lesson deals only with *two-stage* events.

You do not need to know this subject

Everything is worked out by listing outcomes and counting, so a student who can write neatly can do this sheet. Every sample space is printed in full in the key.

Photocopy these pages only

Pages 47–49

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. A pair of dice and a coin are a bonus, not a requirement.

WHAT TO SAY TO START

“One coin has two outcomes and everybody can handle that. Two coins have four, and suddenly half the room thinks getting one of each is as likely as getting two heads. It is not, and finding out why is today.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the two-stage panel	List everything, then count
14–26	Task 9.1 — two coins	Write out all four
26–38	Task 9.2 — two dice	The grid does the work
38–48	Task 9.3 — multiplying along branches	The shortcut
48–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Why are there four outcomes for two coins, not three?”

Because head-then-tail and tail-then-head are different outcomes even though they look the same afterwards. That is exactly why one of each is twice as likely as two heads.

“When do I multiply?”

When you want both things to happen and the first does not change the second. Along a branch of a tree, multiply.

“What changes without replacement?”

The second fraction. If you take a red counter out of five and do not put it back, there are only four left. Task 9.4 does both versions side by side.

If they finish early

Task 9.6 asks whether a tail is “due” after four heads in a row. It is not, and the argument against it is worth having properly — the coin has no memory.

Curriculum focus • for the regular teacher’s records

Grade 7–8: finding probabilities of compound events using organised lists, tables and tree diagrams, and understanding independence. Discussion question for the last five minutes: *“Why is getting one head and one tail more likely than getting two heads?”*

Listing every outcome

UNDERSTAND

12 MIN

NAME _____

CLASS _____

DATE _____

Two things in a row means listing both

TWO COINS

HH HT TH TT

four outcomes, all
equally likely

$$P(\text{two heads}) = \frac{1}{4}$$

$$P(\text{one of each}) = \frac{2}{4}$$

TREE DIAGRAM

```

      H — H   HH
     /  \  L T   HT
start
     \  /  H   TH
      T — T   TT
  
```

follow a branch to the
end for one outcome

MULTIPLY ALONG A BRANCH

$$\begin{aligned}
 P(\text{head and six}) &= \frac{1}{2} \times \frac{1}{6} \\
 &= \frac{1}{12}
 \end{aligned}$$

This works when the first
event does not change
the second.

Task 9.1 • Two coins

DO

a. List all four outcomes of tossing a coin twice.

#	PROBABILITY OF...	ANSWER
b	two heads	
c	two tails	
d	one of each	

#	PROBABILITY OF...	ANSWER
e	at least one head	
f	no heads at all	
g	a head on the second toss	

Dice, branches and bags

KEY SKILL

28 MIN

Task 9.2 • Two dice

DO

Complete the grid of totals, then use it. There are **36** outcomes altogether.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	—	—	—
3	4	—	—	—	—	—
4	5	—	—	—	—	—
5	6	—	—	—	—	—
6	7	—	—	—	—	—

#	PROBABILITY OF...	ANSWER
a	a total of 7	
b	a total of 2	
c	a double	

#	PROBABILITY OF...	ANSWER
d	a total over 9	
e	both dice even	
f	a total of 7 or 11	

Task 9.3 • Multiply along the branch

DO

In each case the first event has no effect on the second, so multiply.

#	THE TWO EVENTS	FIRST	SECOND	BOTH
a	a coin lands heads and a die shows 6			
b	two reds from a spinner that is one-quarter red			
c	rain on both of two days, each 30% likely			
d	a die shows an even number twice in a row			

Task 9.4 • Put it back, or keep it out

COMPARE

A bag holds **3 red** and **2 blue** counters. You take two.

With replacement — the first is put back.

a. $P(\text{first red}) =$ _____

b. $P(\text{second red}) =$ _____

c. $P(\text{both red}) =$ _____

g. Which of the six answers changed, and why only that one? _____

Without replacement — the first is kept out.

d. $P(\text{first red}) =$ _____

e. $P(\text{second red}) =$ _____

f. $P(\text{both red}) =$ _____

Memory, and the long way round

STRETCH

15 MIN

Task 9.5 • Find the error

DETECTIVE

Abe

"P(two heads)
= $1/2 + 1/2$
= 1"

What went wrong?

Correct answer: _____

Bex

"Four heads in a
row, so a tail is
due next toss"

What went wrong?

Correct answer: _____

Cleo

3 red of 5, no
replacement:
" $3/5 \times 3/5$ "

What went wrong?

Correct answer: _____

Task 9.6 • Think it through

CHALLENGE

- a. Which is more likely on two dice, a total of 7 or a total of 11? Say how much more. _____
- b. Find $P(\text{at least one head in two tosses})$ by working out $P(\text{no heads})$ first and subtracting from 1. _____
- c. Three coins are tossed. How many outcomes, and what is $P(\text{all three heads})$? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Chance Twice

ANSWERS

Task 9.1 • Two coins

- a. HH HT TH TT
- b. two heads $\frac{1}{4}$
- c. two tails $\frac{1}{4}$
- d. one of each $\frac{2}{4} = \frac{1}{2}$
- e. at least one head $\frac{3}{4}$
- f. no heads at all $\frac{1}{4}$
- g. head on second $\frac{2}{4} = \frac{1}{2}$

d is the whole lesson. One of each is twice as likely as two heads, because there are two ways to get it and only one way to get HH. Students who list three outcomes rather than four get $\frac{1}{3}$ and cannot see why they are wrong until they write out HT and TH separately.

Task 9.3 • Multiplying along a branch

- a. $\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$
- b. $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$
- c. $0.3 \times 0.3 = 0.09$ (9%)
- d. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

c is worth pausing on: two days each 30% likely give only a 9% chance of both.

Task 9.2 • Two dice • the full grid of totals

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

- a. total 7 $\frac{6}{36} = \frac{1}{6}$
- b. total 2 $\frac{1}{36}$
- c. a double $\frac{6}{36} = \frac{1}{6}$
- d. total over 9 $\frac{6}{36} = \frac{1}{6}$
- e. both even $\frac{9}{36} = \frac{1}{4}$
- f. total 7 or 11 $\frac{8}{36} = \frac{2}{9}$

Seven is the most likely total because more combinations produce it than any other — six of the thirty-six. Two and twelve each have exactly one, which is why they feel so rare at a board game. In d the totals over nine are 10, 11 and 12, which occur three, two and one time each.

Task 9.4 • Put it back, or keep it out

WITH replacement

- a. $\frac{3}{5}$ b. $\frac{3}{5}$ c. $\frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$

WITHOUT replacement

- d. $\frac{3}{5}$ e. $\frac{2}{4}$ f. $\frac{3}{5} \times \frac{2}{4} = \frac{6}{20} = \frac{3}{10}$

g. Only the second draw changed. The first draw happens before anything has been removed, so it is $\frac{3}{5}$ either way. By the second draw there are two reds among four instead of three among five.

Task 9.5 • The three errors

Abe — $\frac{1}{4}$. Adding probabilities is for *or*; multiplying is for *and*. His answer of 1 would mean two heads are certain.

Bex — still $\frac{1}{2}$. The coin has no memory of the four tosses before. This belief has a name, the gambler's fallacy, and it is worth naming out loud because it costs adults real money.

Cleo — $\frac{3}{5} \times \frac{2}{4}$. Without replacement there are only four counters left and two of them are red.

Task 9.6 • Challenge

a. A total of 7, by three times. Seven can happen six ways (1+6, 2+5, 3+4, 4+3, 5+2, 6+1) and eleven only two (5+6, 6+5). So $\frac{6}{36}$ against $\frac{2}{36}$.

b. $\frac{3}{4}$. P(no heads) is P(TT) = $\frac{1}{4}$, so P(at least one) = $1 - \frac{1}{4} = \frac{3}{4}$. It matches Task 9.1e, which is the point — the complement is a shortcut, not a different answer. With ten coins, listing is hopeless and the complement is the only sensible route.

c. Eight outcomes, and P(HHH) = $\frac{1}{8}$. Each extra coin doubles the number of outcomes: 2, 4, 8, 16. Accept $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$ as the working.

Seeing in 3D

2 MINUTES

Students read a solid made of cubes from its plan, front and side views, work out the shape you get when a solid is sliced, and match nets to the solids they fold into. There is almost no arithmetic — this one is about seeing.

You do not need to know this subject

Every answer is a shape name or a small whole number. Nothing needs a formula and nothing needs a calculator, so a sub can mark it from the key at a glance.

Photocopy these pages only

Pages 52–54

One per student. Do not photocopy this page or the answer key.

What students need

A pencil. Any loose cubes or blocks in the room are a bonus.

WHAT TO SAY TO START

“Architects never draw a building in 3D for the builders. They draw it flat, three times, from three directions — and that is enough to build from. Today you read those three drawings and put the solid back together in your head.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the three-views panel	Plan is from above
14–26	Task 10.1 — counting cubes	Slowest part, expect it
26–36	Task 10.2 — slicing solids	Picture the cut face
36–46	Task 10.3 — nets	Count the faces
46–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“What is a plan view?”

The view looking straight down from above. The front view looks from the front, the side view from one side. Between them they describe the solid.

“What is a cross-section?”

The flat shape you see on the cut face when you slice through a solid. Slicing a cylinder across gives a circle; slicing it down the middle gives a rectangle.

“Do three views always describe one solid exactly?”

No, and Task 10.6 is about exactly that. Cubes can hide behind other cubes, so more than one solid can produce the same three drawings.

If they finish early

Task 10.6 asks whether two different solids can give identical plan, front and side views. They can, and finding an example is a genuinely satisfying puzzle.

Curriculum focus • for the regular teacher’s records

Grade 7–8: describing the two-dimensional figures that result from slicing three-dimensional figures, and representing solids with plans and elevations. Discussion question for the last five minutes: “Why do builders work from three flat drawings instead of one picture?”

Reading three views

UNDERSTAND

14 MIN

NAME _____

CLASS _____

DATE _____

Three flat drawings describe a solid

THE THREE VIEWS

PLAN looking down**FRONT** looking at the front**SIDE** looking from the left

Each one is flat. Nothing is drawn at an angle.

HEIGHTS ON A PLAN

How many cubes are stacked on each square:

	left	right
back	2	1
front	3	1

total = $2+1+3+1 = 7$ cubes

SLICING

cylinder cut across
→ **circle**

cylinder cut down the middle → **rectangle**

cone cut across
→ **circle**

Task 10.1 • Reading a solid made of cubes

DO

A solid is built on a 2 by 2 base. The numbers show how many cubes are stacked on each square, seen from above.

	left	right
back	2	1
front	3	1

a. How many cubes altogether? _____

b. How tall is the tallest stack? _____

c. Sketch each view. The front view has two columns; the side view has two columns as well.

PLAN VIEW (FROM ABOVE)

FRONT VIEW

SIDE VIEW (FROM THE LEFT)

Slices, nets and rebuilding

KEY SKILL

24 MIN

Task 10.2 • Slice it

DO

Name the flat shape you would see on the cut face.

#	SOLID	HOW IT IS CUT	SHAPE OF THE CUT FACE
a	cylinder	straight across, parallel to the base	
b	cylinder	straight down through the middle	
c	cone	straight across, parallel to the base	
d	cone	straight down through the point	
e	sphere	any way at all	
f	square pyramid	straight across, parallel to the base	
g	rectangular prism	parallel to one of the faces	

Task 10.3 • Match the net to the solid

DO

#	THE FLAT NET IS MADE OF...	IT FOLDS INTO A...
a	6 squares	
b	2 circles and 1 rectangle	
c	1 square and 4 triangles	
d	2 triangles and 3 rectangles	
e	1 circle and 1 curved sector	
f	4 triangles and nothing else	

Task 10.4 • Build it from the views

REVERSE

A solid is made of cubes on a 2 by 2 base. Its **front view** is two columns, the left 3 tall and the right 2 tall. Its **side view** is two columns, both 2 tall.

a. What is the tallest stack? _____

c. What is the largest number that could? _____

b. What is the smallest number of cubes that could produce these views?

d. Why is there more than one answer?

Task 10.5 • Find the error

DETECTIVE

Gil

"The plan view of a cylinder standing upright is a rectangle"

What went wrong?

Correct answer: _____

Hal

counts a stack of cubes and only includes the ones he can see

What went wrong?

Correct answer: _____

Ivy

"Every slice of a cone gives a triangle"

What went wrong?

Correct answer: _____

Task 10.6 • Think it through

CHALLENGE

- Can you slice a cube so that the cut face is a triangle? Say where you would cut. _____
- Which solid has a circular plan view and a triangular front view? _____
- Can two different solids give exactly the same three views? Explain. _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Seeing in 3D

ANSWERS

Task 10.1 • Reading a solid made of cubes

heights	left	right	a. total = $2+1+3+1 = 7$ cubes
back	2	1	b. tallest stack = 3
front	3	1	
PLAN	FRONT	SIDE (from the left)	
□ □	■	■	
□ □	■ ■	■ ■	
	■ ■	■ ■	
a 2×2 square	3 tall then 1	3 tall then 2	

The plan is simply the 2 by 2 footprint. For the front view, look down each left-right column and take the **tallest** stack in it: the left column has 2 and 3, so it shows 3; the right column has 1 and 1, so it shows 1. For the side view, do the same across each front-back row: the front row has 3 and 1 so it shows 3, the back row has 2 and 1 so it shows 2. Taking the maximum, not the total, is the step students miss.

Task 10.2 • Slice it

a. cylinder across	circle
b. cylinder down middle	rectangle
c. cone across	circle
d. cone through point	triangle
e. sphere any way	circle
f. pyramid across	square
g. prism parallel	rectangle

a and b are the same solid cut two ways, and they give completely different shapes. The sphere is the only solid on the list that gives the same answer however you cut it, which is worth noticing.

Task 10.3 • Match the net

a. 6 squares	cube
b. 2 circles + rectangle	cylinder
c. square + 4 triangles	square pyramid
d. 2 triangles + 3 rects	triangular prism
e. circle + sector	cone
f. 4 triangles	triangular pyramid

c and f both fold into pyramids and the only difference is the base. Counting faces first, then asking what the base must be, settles every row.

Task 10.4 • Build it from the views

front	3 then 2	side	2 and 2
a. tallest stack =	3		
b. fewest cubes =	6		
c. most cubes =	8		

d. Because a taller stack hides a shorter one behind it. The views only record the tallest stack in each line of sight, so anything shorter sitting behind it is invisible from every one of the three directions. That is why builders get plans as well as elevations — three views are not always enough on their own. Accept any answer that names hidden cubes.

Task 10.5 • The three errors

Gil — a circle. Looking straight down on an upright cylinder you see the flat circular top. The rectangle is the *front* view.

Hal — count the hidden ones too. A cube behind another cube still exists. This is the same idea as Task 10.4 seen from the other side.

Ivy — only if the cut goes through the point. A cone sliced parallel to its base gives a circle, as in Task 10.2c.

Task 10.6 • Challenge

a. Yes — slice a corner off. A cut passing through three edges that meet at one corner leaves a triangular face. Any answer describing a diagonal cut across a corner is correct. It surprises students that a shape made entirely of squares can produce a triangle.

b. A cone. Circular from above, triangular from the front.

c. Yes. Take the solid from Task 10.4 with six cubes and add two more behind the tall stack, hidden from every direction — all three views are unchanged. Any example with cubes concealed behind taller ones is full marks, and a student who says “no” and cannot find a counter-example has still thought about it properly.

Thank you for covering this class. Please spend two minutes on this page before you leave and put it on the desk — it is the only handover the regular teacher will get.

SUBSTITUTE'S NAME

DATE

Class / period

Room

WHICH LESSON DID YOU USE?

- ☐ Lesson 01 • Square roots & perfect squares
- ☐ Lesson 03 • Simple & compound interest
- ☐ Lesson 05 • Functions & function rules
- ☐ Lesson 07 • Angles in polygons
- ☐ Lesson 09 • Compound events & tree diagrams

- ☐ Lesson 02 • Rational & irrational numbers
- ☐ Lesson 04 • Similar figures
- ☐ Lesson 06 • Systems of equations
- ☐ Lesson 08 • Scatter plots & correlation
- ☐ Lesson 10 • Cross-sections, plans & elevations

How far did the class get?

- ☐ Finished everything, including the challenge
- ☐ Finished the first two pages
- ☐ Finished the first page only
- ☐ Did not get started — see notes below

The sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Marked together in class using the key

ANYTHING THE TEACHER SHOULD KNOW

STUDENTS WHO WORKED ESPECIALLY WELL

STUDENTS WHO NEEDED MORE SUPPORT

From one teacher to another

Covering a class you do not teach, for students you do not know, from a folder you have never seen is genuinely hard. If today was messy, that is not a reflection on you. Write down what happened, leave the page here, and go home. Thank you.

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Standards alignment

Written against the Grade 6–8 band of the Common Core State Standards for Mathematics, and readily mapped to state equivalents and to the Australian Curriculum Years 7–8.

- **01 • The Root of It** — Square roots & perfect squares
- **02 • Never Ending** — Rational & irrational numbers
- **03 • Money Over Time** — Simple & compound interest
- **04 • Same Shape, Different Size** — Similar figures
- **05 • In and Out** — Functions & function rules
- **06 • Two Lines, One Answer** — Systems of equations
- **07 • Many Sides** — Angles in polygons
- **08 • Scattered** — Scatter plots & correlation
- **09 • Chance Twice** — Compound events & tree diagrams
- **10 • Seeing in 3D** — Cross-sections, plans & elevations

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